Capstone Project (Classification Flavor)

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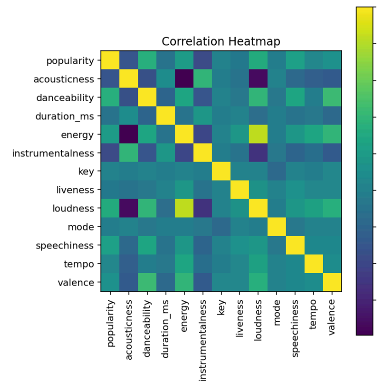
AI Usage: I used Google Gemini to resolve warnings and errors and to make sure that my code produces the desired results. For example, my prompts were, "Does this code correctly perform PCA with two components and color the projected solution using different colors based on the true label," and "How to resolve this warning about early stopping before convergence?"

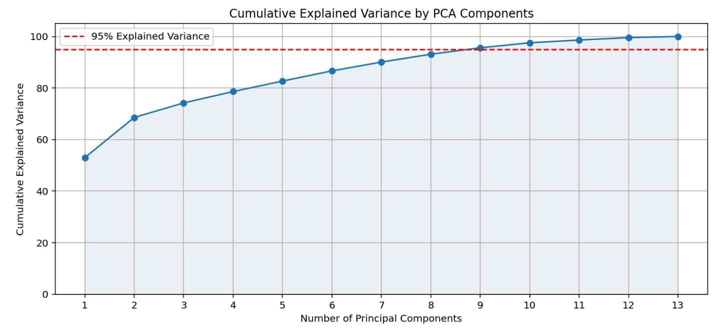
Preprocessing: After loading in the dataset using pandas, I immediately dropped the linguistic variables "instance_id", "artist_name", "track_name", and "obtained_date" from the dataset since they have little predictive power of the music genre of a track and would be hard to encode into numeric labels. I then checked for missing values and realized that other than the 5 entirely empty observations, the features "tempo" and "duration_ms" each had almost 5,000 values of "?" or -1.0, which are different from the other continuous values and impossible to be true. Due to the large numbers of missing values, instead of dropping all those observations, I filled each observation in with the median value, rather than the mean since these features are not normally distributed, of "tempo" or "duration_ms" of the corresponding music genre. After that, I handled categorical variables "key" and "mode" by mapping each distinct string value to a numeric value, and the categorical outcome variable "music_genre" using a label encoder. To perform a train/test split using strictly 500 randomly selected songs from each genre in the test set, I set the test set size to 5000 and stratified on "music_genre", resulting in a training set with 45,000 observations and a test set with 5,000 observations. Since PCA and neural networks are highly sensitive to the unit of features, I normalized the continuous features and concatenated them with the unnormalized categorical variables to form the final training and test sets.

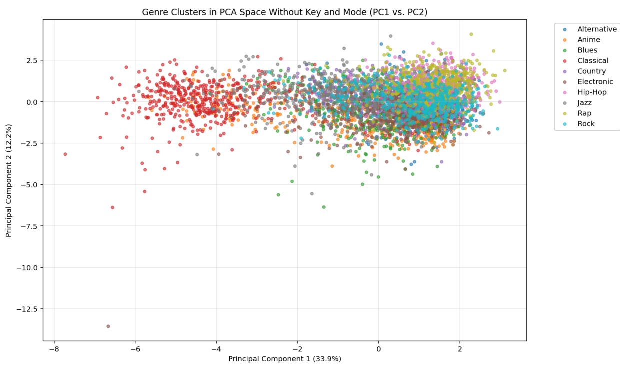
What I Did: After preprocessing the music dataset, I plotted a correlation heat map to inspect any collinearity in the music dataset. I then fitted a PCA model using the scaled training set and kept the top nine principal components that cumulatively explain 95% of variance in the features. To visualize the genres as cluster in a lower dimensional space, I made two scatter plots each using a subset of 5,000 points and the first two principal components of two new PCA models, one with all features and the other with only the numeric features to allow for both the stripe-shaped clusters caused by the categorical feature "key" and the continuous data spread that better shows how the auditory features group together. Additionally, I plotted the loadings of the two PCA models to analyze the contributions of each feature in maximizing variance in the first two orthogonal dimensions. As a clustering step, I performed two K-Means models with k equals 10 on each of the two PCA models to examine whether there's any inherent groupings in the music dataset. Finally, I fitted a random forest model, a gradient boosting model, and a neural network to dynamically classify music genres, and plotted the ROC and PR curves to visualize classification success across different classification models.

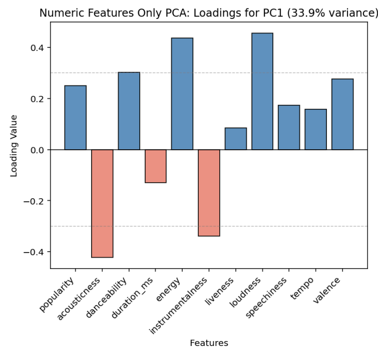
Design Choices: To mitigate the non-normality of acoustic features, I utilized a standard scaler to center the data and ensure that variance-based PCA was not biased by feature magnitude. Additionally, applying dimensionality reduction before clustering or fitting a classification model also allows feature extraction, which reduces collinearity between correlated features, like "energy," "loudness," and "acousticness," as observed in the correlation heat map, as well as the risk of overfitting to noise in high dimensions. Moreover, applying clustering to dimensionality-reduced data allows us to analyze the relationships and any natural groupings of the tracks in a lower dimension where distances are meaningful. According to the overlapping genre clusters in the PCA solution and the low silhouette score of the K-Means solution (0.163), this music dataset exhibits nonlinear relationships between features. Therefore, in choosing the best model for multi-class classification, I prioritized random forests, gradient boosting, and neural networks not only because of their ability to handle the non-normal features in this dataset but also the complex, nonlinear interactions between them. In crafting the models, I designed the random forest model to build 200 trees with random feature selection at each node so that the model wouldn't overfit to noise; I crafted the gradient boosting model to have 500 trees with a learning rate of 0.1 to allow enough room for sequential learning and fast convergence in a small dataset; I made the neural network to have a logistic activation with 3 hidden layers, maximum number of iterations of 400, and a small learning rate of 0.001 to make sure that the deep network escapes local minima and converges at a global one. Moreover, to prevent overfitting in the deep multi-layer perceptron architecture, I enabled early stopping, which allowed the model to terminate once the validation AUC plateaued, ensuring the sweet spot of generalization was reached. Lastly, for multi-class ROC curve and PR curve plotting, I first binarized the output labels, treating only the true label as the correct one and the other nine labels as incorrect for every case. I then calculated each model's AUC using the dimension-reduced test set and the multi-class evaluation function of "roc_auc_score" and plotted both the ROC and PR curves to show how well the models distinguish true positives from false as well as how precise their predictions are.

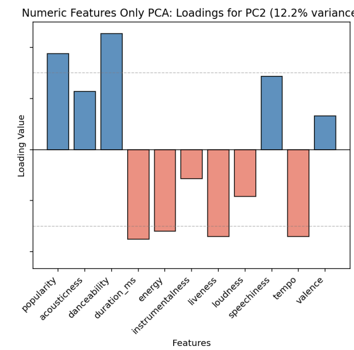
Results: To retain 95% of variance in the dataset, the first nine principal components are required. According to the colored PCA solution without categorical features, the music genres are roughly divided into two clusters, one consisting of the classical tracks and the other all the rest. The loadings suggest that PC1, which explains 33.9% of variance, captures the inverse relationship between acousticness and energy, and data points with low PC1 values have low energy and high acousticness, explaining the classical music cluster dominating the left half of the PCA projection. PC2, which explains 12.2% of variance, likely represents the atmosphere of a track, where a data point high up on the scatter plot would be more danceable and popular, explaining why rap music primarily sits on the higher end of PC2 while jazz music sits at the lower end. In the 2D PCA projection with all features, where there are twelve strip-shaped clusters each corresponding to one of the twelve keys, classical tracks are also separated from all other ones along principal component two. Closely inspecting the loadings, we can see that PC1 is dominated by the feature “key” with near zero contribution from other features and explains 53% of variance, 20% higher than that of PC1 in the PCA model with only numeric features, which PC2 closely resembles in terms of loadings, meaning that the classical tracks still differ from all other tracks in acoustic features, regardless of key. Looking at the K-Means solution on numeric features only, the overlapping cluster centroids prove that the algorithm was unable to find distinct clusters in the PCA solution, indicating that there is nonlinearity in this dataset that cannot be mapped on a lower dimension using a linear transformation. On the other hand, the neural network, gradient boosting model, and random forest each yielded AUC’s of 0.923, 0.904, and 0.896, and average precision scores of 0.598, 0.528, and 0.499. When using not only the first nine PC’s but also the the clusters yielded by K-Means, the AUC’s decreased to 0.922, 0.904, and 0.893, meaning that the clusters found by K-Means are not representative of the true groupings of the music tracks, and previous test runs proved that using them in the classification models only makes them worse, which is why I excluded the clusters and used the PCA solution only when fitting the final classification models. From the ROC and PR curves, we can see that the multi-layer perceptron achieved both the highest AUC and average precision score, making it the best classifier of music genre that provides the most accurate and precise predictions.

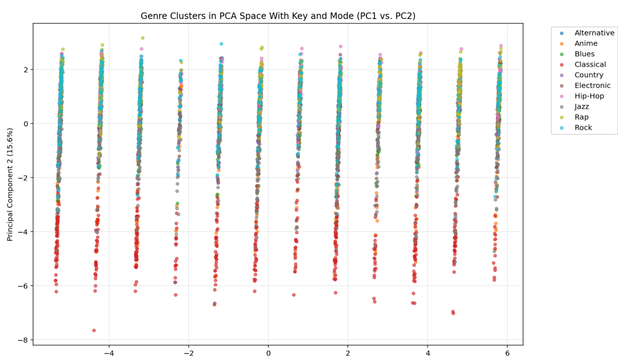

Correlation Heatmap
popularity
acousticness
danceability
duration_ms
energy
instrumentalness
key
liveness
loudness
mode
speechiness
tempo
valence
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

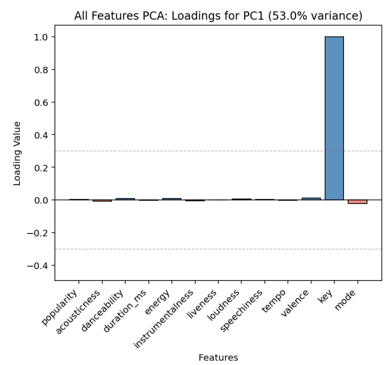

Cumulative Explained Variance by PCA Components
95% Explained Variance
Cumulative Explained Variance
Number of Principal Components

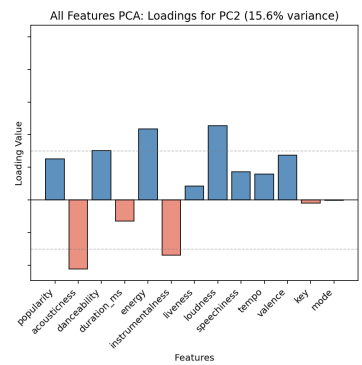

Genre Clusters in PCA Space Without Key and Mode (PC1 vs. PC2)
Alternative
Anime
Blues
Classical
Country
Electronic
Hip-Hop
Jazz
Rap
Rock
Principal Component 1 (33.9%)
Principal Component 2 (12.2%)

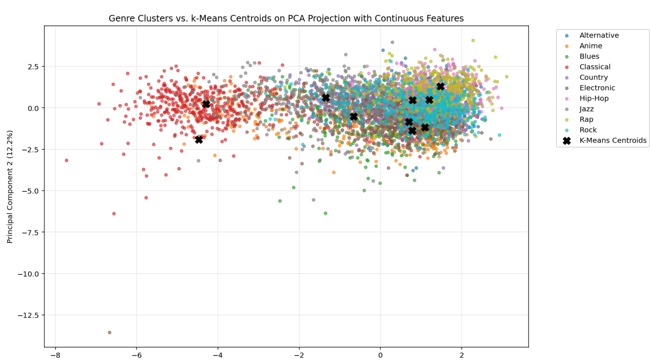

Numeric Features Only PCA: Loadings for PC1 (33.9% variance)
Loading Value
Features
popularity
acousticness
danceability
duration_ms
energy
instrumentalness
liveness
loudness
speechiness
tempo
valence

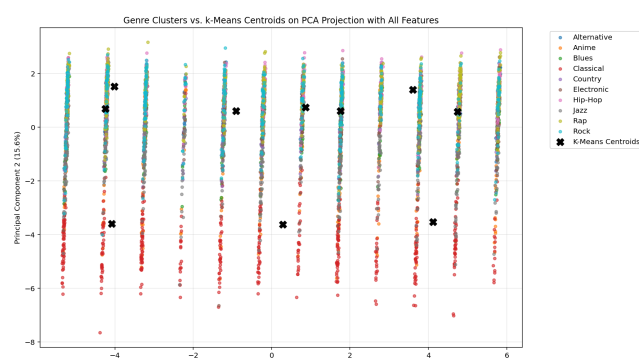

Numeric Features Only PCA: Loadings for PC2 (12.2% variance)
Loading Value
Features
popularity
acousticness
danceability
duration_ms
energy
instrumentalness
liveness
loudness
speechiness
tempo
valence

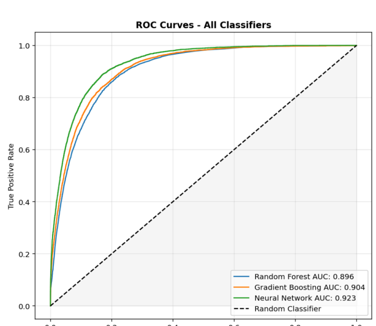

Genre Clusters in PCA Space With Key and Mode (PC1 vs. PC2)
Alternative
Anime
Blues
Classical
Country
Electronic
Hip-Hop
Jazz
Rap
Rock
Principal Component 1 (53.0%)
Principal Component 2 (15.6%)

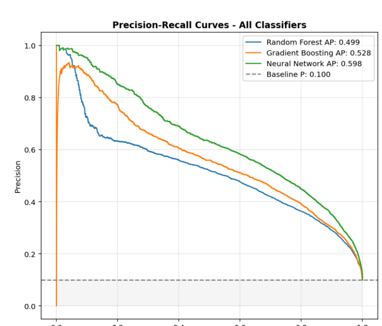

All Features PCA: Loadings for PC1 (53.0% variance)
Loading Value
Features
popularity
acousticness
danceability
duration_ms
energy
instrumentalness
liveness
loudness
speechiness
tempo
valence
key
mode


All Features PCA: Loadings for PC2 (15.6% variance)
Loading Value
Features
popularity
acousticness
danceability
duration_ms
energy
instrumentalness
liveness
loudness
speechiness
tempo
valence
key
mode


Genre Clusters vs. k-Means Centroids on PCA Projection with Continuous Features
Alternative
Anime
Blues
Classical
Country
Electronic
Hip-Hop
Jazz
Rap
Rock
K-Means Centroids
Principal Component 1 (33.9%)
Principal Component 2 (12.2%)


Genre Clusters vs. k-Means Centroids on PCA Projection with All Features
Alternative
Anime
Blues
Classical
Country
Electronic
Hip-Hop
Jazz
Rap
Rock
K-Means Centroids
Principal Component 1 (53.0%)
Principal Component 2 (15.6%)


ROC Curves - All Classifiers
Random Forest AUC: 0.896
Gradient Boosting AUC: 0.904
Neural Network AUC: 0.923
Random Classifier
False Positive Rate
True Positive Rate


Precision-Recall Curves - All Classifiers
Random Forest AP: 0.499
Gradient Boosting AP: 0.528
Neural Network AP: 0.598
Baseline P: 0.100
Recall
Precision

Interpretation: **The neural network, gradient boosting model, and random forest each generated AUC's of 0.923, 0.904, and 0.896, with neural network being the best classifier of the three.** This could be due to the fact that deep neural networks with well-tuned hyperparameters are excellent at finding non-linear patterns, compared to gradient boosting, which is prone to overfitting, and random forests, which can struggle with boundaries between classes, especially when there are many classes. The heatmap revealed high multicollinearity between "energy," "loudness," and "acousticness," and PCA successfully decorrelated the feature space, making it easier for the neural network to achieve a high AUC as it didn't need to navigate redundant and conflicting signals. Additionally, the fact that we need to keep the top nine orthogonal principal components in order to retain 95% of the variance means that most features in this music dataset are relatively independent of each other and carry distinct information about music genre, suggesting that there are non-linear relationships between auditory features and music genre that need many features to uncover altogether, making neural networks the most qualified to handle since they are very flexible and turing-complete, especially when given clean data. Looking at the visualization and loadings of the 2D PCA model with all features, the fact that the feature "key" alone explains about 53% of variance in the dataset suggests that all keys are equally used in all music genres, and no particular key is unpopular. At each key cluster along the second principal component, classical tracks dominate the lower part of the cluster while tracks from almost all other genres cluster together around the top. This substantiates the fact that classical music generally has low energy and loudness but high acousticness and instrumentalness, differentiating itself from the other contemporary genres, which generally exhibits high energy. Furthermore, the PCA visualization with only the numeric features spread continuously along the first principal component, rather than stripe-like, with the classical tracks dominating the left half of the data cloud and the rest of the tracks clustering around the right half, further substantiating the fact that “key” is an important predictor of music genre, and there is inherent overlap in acoustic features between contemporary genres. From the ROC curves, we can see that while all three models have AUC’s much higher than the baseline of 0.5, the deep neural network has the highest discriminative power of 0.923. Moreover, even though every class in the dataset has exactly 5,000 observations, multi-class evaluation requires using the “one versus rest” strategy where only one class is the correct one while the other nine are incorrect, revealing a prevalence baseline of 0.1 rather than 0.5 like in binary classication, further confirming the classification success in the neural network. Furthermore, the PR-curve proves that the neural network not only distinguishes true and false positives well, but also produces the most reliable positive class predictions out of all models since the average precision scores are nearly six times the baseline of 0.1, indicating its excellence at predicting the single correct label out of ten possible ones while handling class imbalance and feature overlap. In conclusion, I think the most important factor that underlies my classification success is model choice. In a dataset with many features, neural networks are superior at uncovering complex, nonlinear relationships since they can approximate any function by iteratively minimizing errors and adjusting millions of weights and biases during backpropagation. Uncorrelated features is also an important factor of classification success, since without it, the models could be confused by the overlapping information between predictor variables. Another important factor would be domain-specific data cleaning, where I filled in the “NaN” values with the median value of the feature grouped by the corresponding music genre of each row, which allowed the observations with missing values to be consistent with the rest of the dataset and prevented the model from experiencing class imbalance.

Additional Observations (Extra Credit)

1. In crafting the neural network, the mutli-layer neural network achieved its highest AUC of 0.925 at 200 iterations. Although the model has not converged according to the solver, empirical testing showed that forcing convergence via increased iterations and lowered learning rates resulted in performance decay, where the best AUC when the model converged was 0.923. This indicates that a non-converged state represents a superior point of generalization, effectively avoiding the overfitting that occurred in extended training.
2. The slight dip of the PR curves below the baseline in the final recall stages indicates that the "tail" of the genre distributions contains highly ambiguous samples. Forcing the model to achieve total recall on these outliers introduces a disproportionate number of false positives, highlighting the inherent boundary overlap between music genres.
3. In the K-Means clustering of the PCA projection with all features, including the categorical ones, the centroids were not evenly distributed across the stripes, and some of them even land in between clusters. This is because K-Means assumes that data came from isotropic Gaussian distributions, so it is only able to yield spherical clusters and wasn't able to identify the strip-like clusters in the PCA projection. Moreover, the fact that centroids could land between clusters shows that they don't need to be real data points, making them not very meaningful and difficult to interpret.